

NEET Important Chapter – Mechanical Properties of Fluids

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Mechanical Properties of Fluids is an important section as far as NEET is concerned. In this chapter we covered all mechanical properties and characteristics of fluid (liquid and gasses) and discussed how fluids are different from solids.

Fluid mechanics are considered as two types. First is fluid statics in which fluid or liquid is stationary (not moving). So in this part we have studied pressure, variation of pressure with depth or height and buoyancy force.

Another part is fluid dynamics in which we have studied about the dynamic nature of fluid and in this section is covered continuity equation, bernoulli equation etc.

The problems with continuity equations, Bernoulli equations are of frequently asked for the NEET exam.

In the later we also discussed the real fluid and their characteristics like viscosity, surface tension and excess pressure and also understand the important concept of surface energy. Let us see what topic and concept of mechanical properties of fluids are important for the NEET Exam with solved examples.

Mechanical Properties of Fluids – Important Topics

- Mechanical properties of fluids notes
- Pascal's Law
- mechanical properties of fluids formulas
- Continuity Equation
- Bernoulli's Principle
- mechanical properties of fluids solutions
- properties of fluid – Viscosity
- Stokes Law
- Surface Energy

Important Concepts of Mechanical Properties of Fluids

Name of the Concept	Key Points of the Concept
1. Fluid Pressure	• The pressure exerted by fluid is known as fluid pressure. And it is defined as a normal force acting on per unit volume of fluid. • Fluid pressure is the same at two points in the same horizontal level. • S.I. unit – — • It is a scalar quantity.

2. Pascal's law	• The pressure exerted at a point is same in all direction in the fluid at rest • Liquid exerts equal pressure in all directions
3. Type of pressure	• Gauge pressure • Absolute pressure • Atmospheric pressure
4. Archimedes principle	• Whenever a body is fully or partially submerged in liquid it experiences a net upward force which is equal to the weight of the liquid displaced.
5. Buoyancy force	• Whenever an object is partially or fully immersed in a fluid or liquid, then it experiences a net upward force due to the surrounding fluid. This phenomenon is called buoyancy and this force is called upthrust force or buoyant force .
6. Relative density	• The relative density of a liquid or fluid is the ratio of its density to the density of water. • It is a scalar quantity and dimensionless
7. Bernoulli theorem	• In fluid dynamics the sum of pressure energy per unit volume , kinetic energy per unit volume and potential energy per unit volume will always remain constant.

8. Viscosity	• Viscosity is defined as the internal resistance offered by a layer of fluid to the adjacent layer during the fluid flow. • M.K. S. Unit of Coefficient of viscosity - —
9. Surface tension	• Surface tension is the property of liquid surface to contract or compress into the minimum surface area possible at rest • It is force per unit length. • It is a tensor quantity. • Unit - —
10. Surface energy	• It is the potential energy contained by the molecules of the liquid surface due to which they stay on the surface against inward force of attraction by other molecules.
11. Streamline motion	• When liquid is in motion, then fluid particles follow the same path or a way, then this motion is called streamline motion and the path is called streamline.
12. Excess pressure	• Due to property of surface tension a drop or a bubble tries to contract and so compress the matter and close first in turn increases the internal pressure is known as excess pressure

List of Important Formulae

Sl. No	Name of the Concept	Formula
1.	Pressure at h depth	
2.	Buoyancy force	Here ρ is density of liquid g is acceleration due to gravity V is Volume of unmerged body or volume of displaced water
3.	Equation of continuity	Here A is the area and v is the velocity
4.	Bernoulli Equation	— Here ρ is density of liquid g is acceleration due to gravity h is height and v is the velocity

5.	Velocity of efflux	It is the velocity of liquid that will come out of a hole at h height from the bottom of the container filled with liquid $v = \sqrt{2gh}$ Here h is the height of hole from top.
6.	Newton' law of viscosity (viscous force)	$F = \eta A \frac{dv}{dy}$ Here η i viscosity
7.	Drag force or viscous drag force	Here η is viscosity , v is velocity of object (spherical) in the liquid and r is radius of spherical object $F_d = 6\pi\eta rv$
8.	Terminal velocity	$v_t = \sqrt{\frac{2r^2(\rho - \sigma)g}{9\eta}}$ Here ρ is density of object and σ is density of liquid

9.	Surface energy	Surface energy = surface tension $\times$ Area
10.	Excess pressure inside a liquid drop	$\frac{4S}{r}$ Here S is surface tension and r is radius
11.	Excess pressure inside a soap bubble	$\frac{8S}{r}$ Here S is surface tension and r is radius
12.	Renold number	$\frac{\rho v D}{\eta}$ Here D is diameter of pipe and v is velocity
13.	Capillary rise (height rise and height fall)	$h = \frac{2S \cos \theta}{r \rho g}$ Here ρ is density of liquid θ is contact angle S is surface tension r is radius of capillary tube

Solved Examples

1. A 100 kg girl wearing high heel shoes balances on a single heel. The heel is circular with a diameter of 1.0 cm. What is the pressure exerted by the heel on the horizontal floor?

Ans:

Given mass of the girl = 100 kg

The diameter of the heel is $d = 1 \text{ cm}$

The radius of the heel is $= d/2 = 0.005 \text{ m}$

Now find area of the heel = $A =$

$A =$ $=$

Due to heel the force on the floor would be $F = mg =$ $= 980 \text{ N}$

Pressure exerted by the heel on the floor by formula -

—

—

For determining pressure find the normal force acting on the surface then divide it by area of surface.

2. Torricelli's barometer used mercury. Pascal duplicated it using French wine of density 984 kg m^{-3} . Determine the height of the wine column for normal atmospheric pressure?

Ans:

We know that the density of mercury,

Density of French wine

We know that height of the mercury column, $= 76 \text{ mm} = 0.76 \text{ m}$

Let consider the height of the wine column =

We know that Acceleration due to gravity, $g = 9.8 \text{ m/s}^2$ —

Pressure in both column mercury and wine is equal so

Key Point: Since both the liquid are at same atmospheric pressure therefore pressure remains the same for both the liquid.

Previous Year Questions from NEET Paper

1. A capillary tube of radius r is immersed in water and water rises in it to a height at the mass of the water in the capillary is 5 gram another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is (NEET 2020)

Ans:

Given - radius of capillary tube is - r

Mass of the water in the capillary is - 5 gm

New radius of capillary tube is - $2r$

Change in pressure -

Multiply the whole equation by

Here ρ is density of liquid (water), g is acceleration due to gravity, r is radius of tube and V is the volume.

By the formula of density

$$M = \rho V$$

Therefore M is proportional to the radius of the tube.

The mass of water that will rise in this tube is two times of 5 = 10 gm

Trick - In this type of question try to make the relation between given and required physical quantity.

2. A small hole of an area of cross-section _____ is present near the bottom of a fully filled open tank of height 2 m. Taking $g=10 \text{ m/s}^2$, the rate of flow of water through the open hole would be nearly:

Ans:

Given - Area of cross section = A = _____

The rate of flow = $Q = Av$

Velocity = $v =$ _____

= $Q =$ _____

= _____

= _____

= _____

Trick - remember all the important formulas and see what is given and what we have to find then apply the relevant formula. In this question, think about the continuity equation and formula of rate of flow of fluid.

Practice Questions

1. Water falls down from a water tap with speed 3 m/s from 2 area then find area of streamline line when water goes down 80 cm (Ans: 1.2)
2. Pressure of water in a pipe when tap is open is — and it become — when it is closed then what is velocity of flow when tap was open? (Ans: 10 m/s)

Conclusion

In this article, we discussed important topics , important Mechanical Properties of Fluid formulas from the NEET point of view. Students must make sure that they do not miss any of the above important topics to obtain a good score in the NEET exam.

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